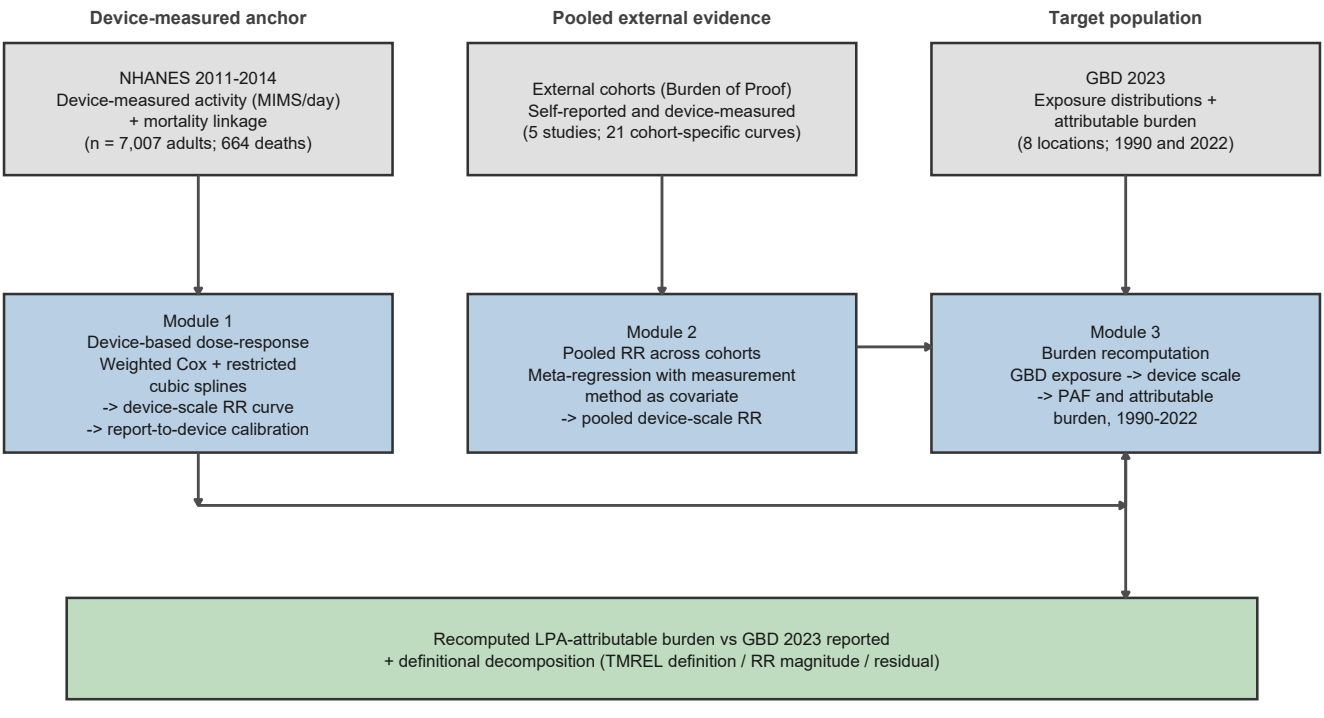
Device-measured anchor
Pooled external evidence
Target population
NHANES 2011-2014
Device-measured activity (MIMS/day)
+ mortality linkage
(n = 7,007 adults; 664 deaths)
External cohorts (Burden of Proof)
Self-reported and device-measured
(5 studies; 21 cohort-specific curves)
GBD 2023
Exposure distributions +
attributable burden
(8 locations; 1990 and 2022)
Module 1
Device-based dose-response
Weighted Cox + restricted
cubic splines
-> device-scale RR curve
-> report-to-device calibration
Module 2
Pooled RR across cohorts
Meta-regression with measurement
method as covariate
-> pooled device-scale RR
Module 3
Burden recomputation
GBD exposure -> device scale
-> PAF and attributable
burden, 1990-2022
Recomputed LPA-attributable burden vs GBD 2023 reported
+ definitional decomposition (TMREL definition / RR magnitude / residual)